

Petit a Petit

Science

*Computation
Clinical Approaches
Epidemiology*

*Statistics
Public Health
Virology*



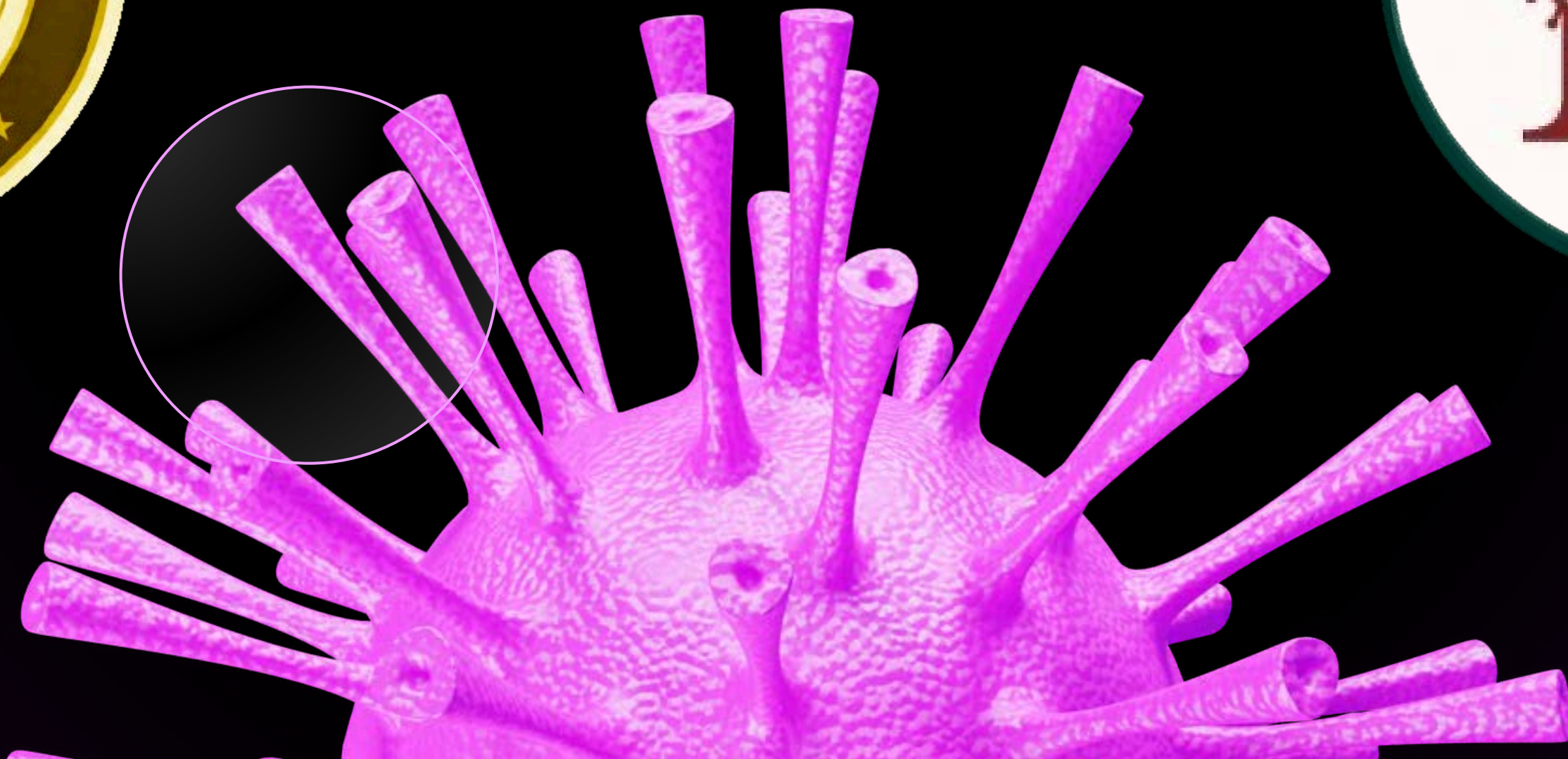
Workshop

Presented by:
Daniel, Mahathi, Sushank, &
Anvika

Follow our socials!

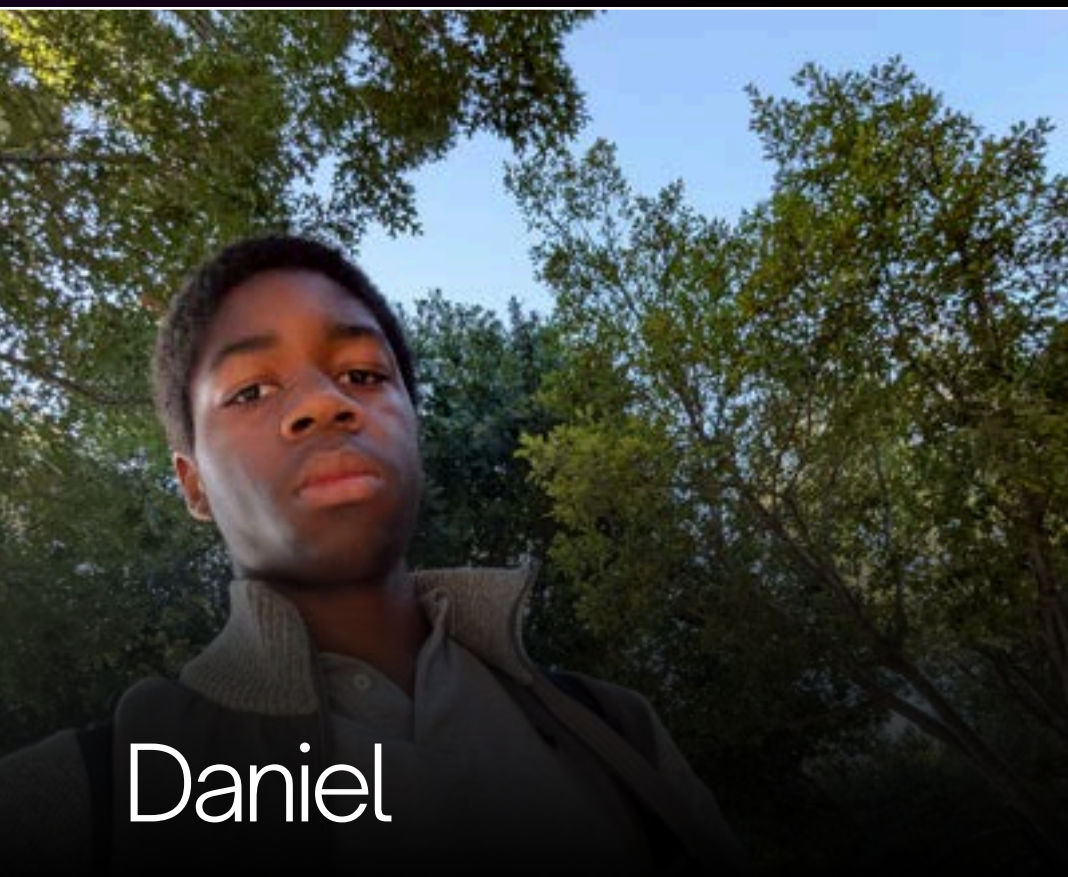


@yimocomp_
@harvardmicrosociety



Our Origins

Welcome everyone! We are representing the **Harvard Undergraduate Microbiology Society (HUMS)** and our NGO, **Petit a Petit**. Our journey began with a shared passion for spreading global awareness about microbiology, which ultimately inspired our team, Sushank, Daniel, Anvika, and Mahathi, to launch our own organization.



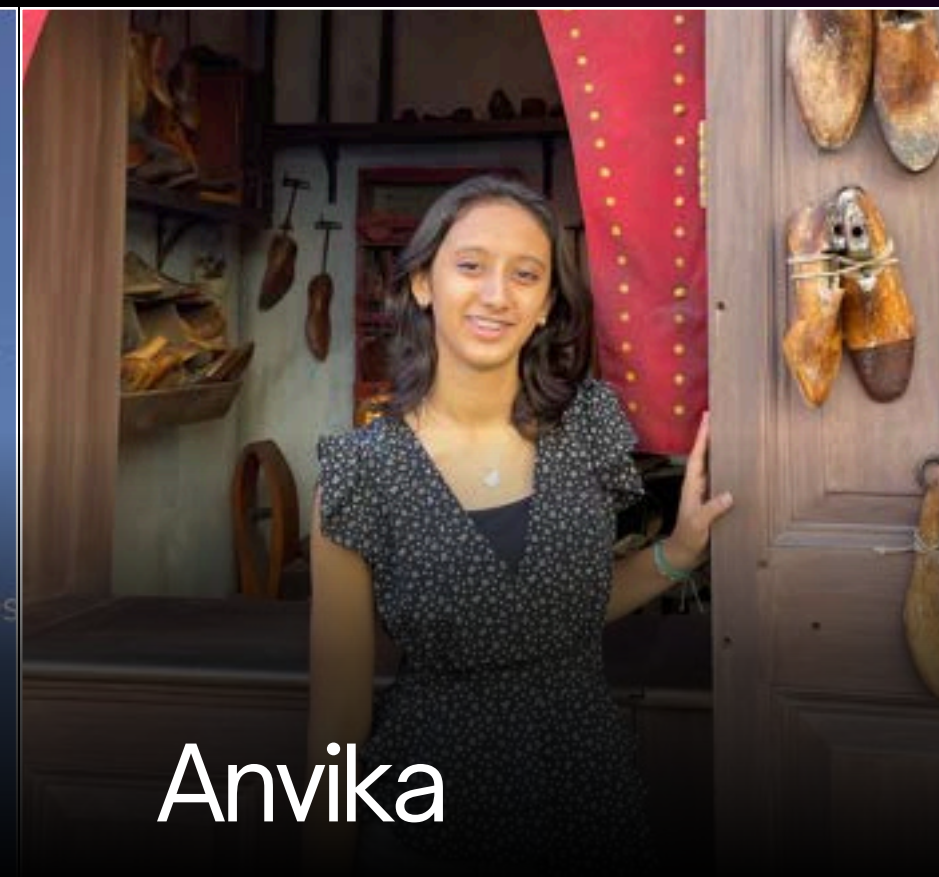
Daniel



Mahathi



Sushank



Anvika



Our Origins

We chose the name Petit a Petit because it means "little by little," reflecting our belief that long-lasting global impact is built one small step at a time. Our ultimate goal as HUMS and Petit a Petit here today, spanning across Europe, Asia, and the Americas, is to spark a deep love for the sciences in young minds like yours and show you how mathematical tools can be used to solve real-world health crises.



The Evolution of Health and Math

Welcome to the crossroads of math and life-saving sciences. This seminar highlights how mathematical tools are the ultimate weapons against global crises.

You have two distinct academic and humanitarian routes to choose from.



Computation & Epidemiology

Modeling the future to Stop the next Pandemic



- *The Microscopic Science: This path begins in the lab, studying the biological behavior, genetic sequencing, and mutation rates of microscopic pathogens.*
- *The Computational Math: We transform biology into data using predictive algorithmic modeling, discrete mathematics, and network theory.*
- *How it Works: By programming digital simulations of human movement and virus behavior, we can map out transmission networks across continents.*
- *The Ultimate Goal: This route acts like a digital weather forecast. Instead of predicting rain, your algorithms simulate outbreaks before they happen, allowing global health systems to intercept a pandemic before it starts.*

Statistics & Public Health

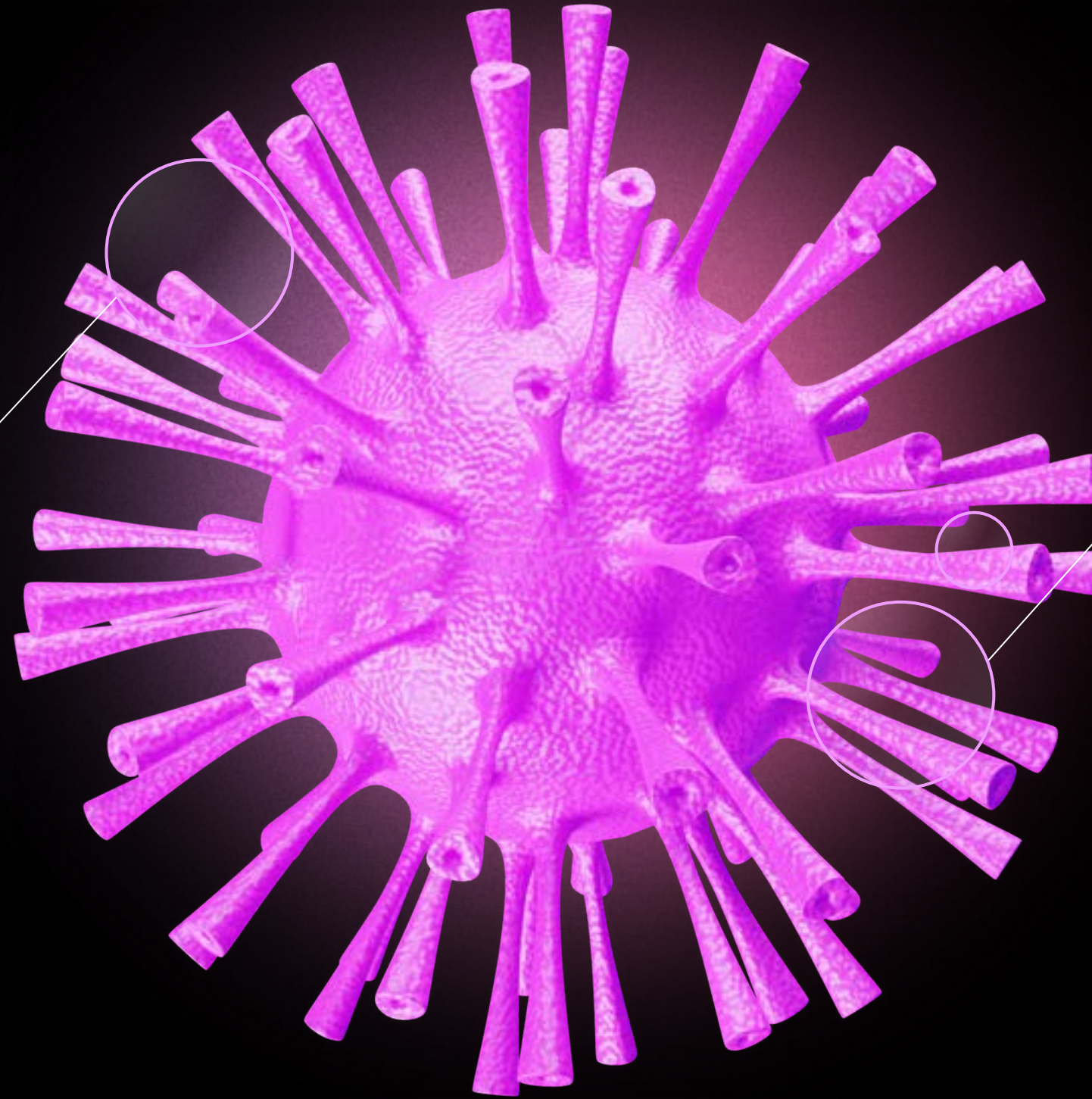
Analyzing the Present to Protect Communities Today



- *The Real-World Science: This path **operates on the ground**, analyzing how environmental factors, local infrastructure, and human behaviors impact community health.*
- *The Statistical Math: The process of messy, real-world data **using biostatistics, probability distributions, and big data regression models**.*
- *How it Works: Instead of simulations, we **look at actual case counts, geographical clusters, and population demographics** to find hidden patterns of disease spread.*
- *The Ultimate Goal: This route creates the shield that protects humanity. Statistical evidence **directly proves which medical interventions work**, which helps global health systems to deploy limited medical resources exactly where they will save the most lives right now.*

Why Are Viruses So Small?

Surface Area $\propto r^2$
Volume $\propto r^3$



As size decreases:

- Surface area decreases more slowly than volume
- Surface-area-to-volume ratio increases
- Molecular interactions become more efficient

-
- *Faster Diffusion*
 - *Efficient Receptor Binding*
 - *Compact Genome Packaging*
 - *Rapid Interaction w/ Host Cells*

Viral Genome Architecture & Replication Strategy

Viral Genome Types

dsDNA Viruses

- Found in the nucleus
- Use host’s DNA/RNA polymerase
- More stable genetically

SSRNA(+) Viruses

- Found in the cytoplasm
- Use their own RNA-dependent RNA polymerase
- Translated immediately by host ribosomes

SSRNA(-) Viruses

- Carry their own RNA-dependent RNA polymerase
- Convert negative sense RNA to positive sense RNA
- Must rewrite genome before making proteins

Retroviruses

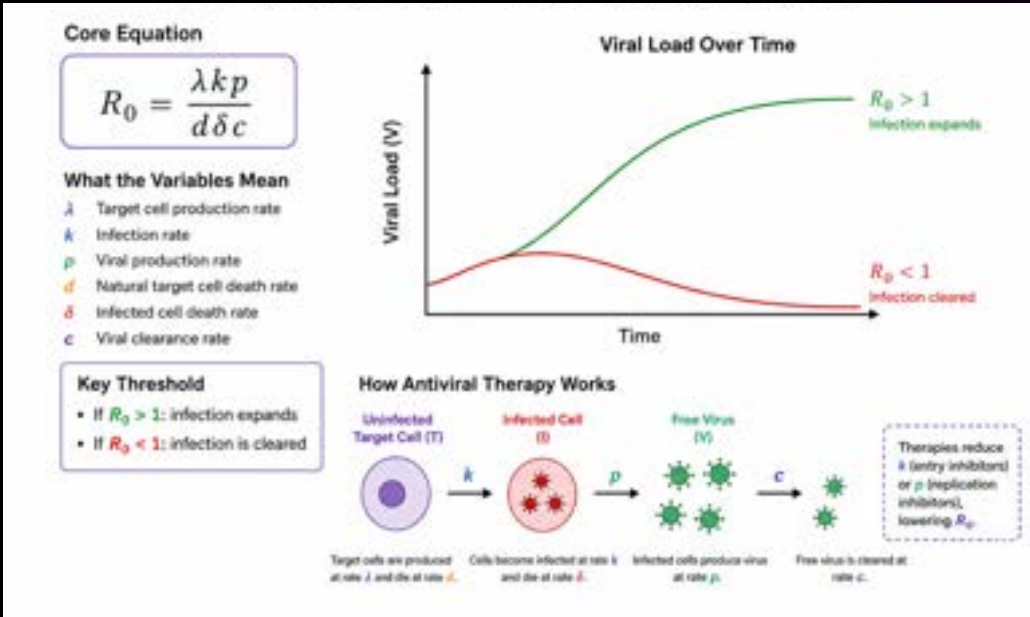
- Use reverse transcription (RNA → DNA)
- Viral DNA becomes part of host genome

Key Molecular Differences

- Replication Location: nucleus vs cytoplasm
- Dependence on host polymerase
- Genome segmentation
- Reverse transcription

Group	Group Name	mRNA generation mechanism
I	dsDNA	Transcription takes place directly from ds DNA.
II	ssDNA	DNA is made ds by replication before the transcription.
III	dsRNA	mRNA is produced from genomic RNA
IV	ssRNA (+)	Replication to generate the negative strand from positive strand. Negative strand is transcribed
V	ssRNA (-)	Negative strand is directly transcribed.
VI	RT ssRNA	RT is used to generate DNA from genomic RNA. This DNA is used for transcription.
VII	RT dsDNA	DNA is transcribed to generate mRNA. This mRNA is reverse transcribed to generate DNA.

Research Gate: Baltimore Classification System



**R naught describes how efficiently infection spreads between cells inside a host

Host Cell Hijacking & Immune Evasion

Ribosome Hijacking:

- Normalcy: ribosomes translate host mRNA into cellular proteins
- Change: viruses redirect ribosomes toward viral RNA (instead of host RNA)
- Ex: Poliovirus

Cap-Snatching:

- Normalcy: eukaryotic mRNA has a 5' cap (stability, ribosome binding)
- Change: viruses steal caps from host mRNA and attach them to viral RNA
- Ex: Influenza

Polyprotein Cleavage:

- Normalcy: individual production of proteins by cells
- Change: RNA viruses produce a giant polyprotein which is later cleaved into functional viral proteins
- Ex: Poliovirus and many flaviviruses

Membrane Remodeling:

- Normalcy: cellular membranes organize organelles
- Change: viruses remodel membranes into compartments which shield viral RNA from immune sensors
- Ex: COVID-19

Interferon Suppression:

- Normalcy: infected cells release interferons; antiviral signaling molecules and activate neighboring immune defenses
- Change: blocks interferon production and pathway signaling; Ex: some coronaviruses

Latency:

- Normalcy: viral replication triggers immune detection
- Change: viruses enter a dormant state with minimal gene expression
- Ex: Herpes

Genome Integration:

- Normalcy: host DNA remains separate from foreign genetic material
- Change: retroviruses integrate viral DNA into host genome
- Ex: HIV

Evolution & Mutation Rates

- As mutation rate increases, evolution speed increases
- RNA polymerases error-prone → rapid mutation → quasispecies: genetically diverse viral populations

Biology in Relation to Statistics



Statistics in the field of biology is based off of using math and data analysis to understand biological systems. While biology has an extremely important role in virology and epidemiology, statistics plays a role in determining factors such as

- how each person responds to a disease
- how genes behave in different environments
- how virus transmissions change between populations

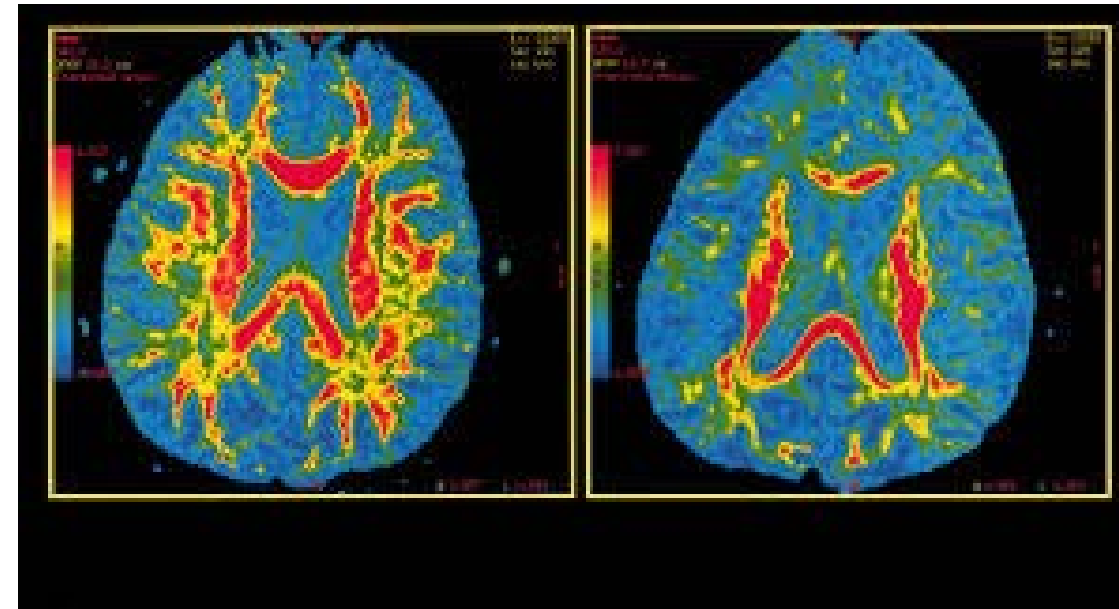
Where does Computational Biology tie into Statistics?

Computational Biology uses computers to process massive datasets while statistics provides math tools to analyze that data. Statistics in biology is a continuation of using previous principles like

- variables,
- probability,
- correlation vs. causation,
- sampling

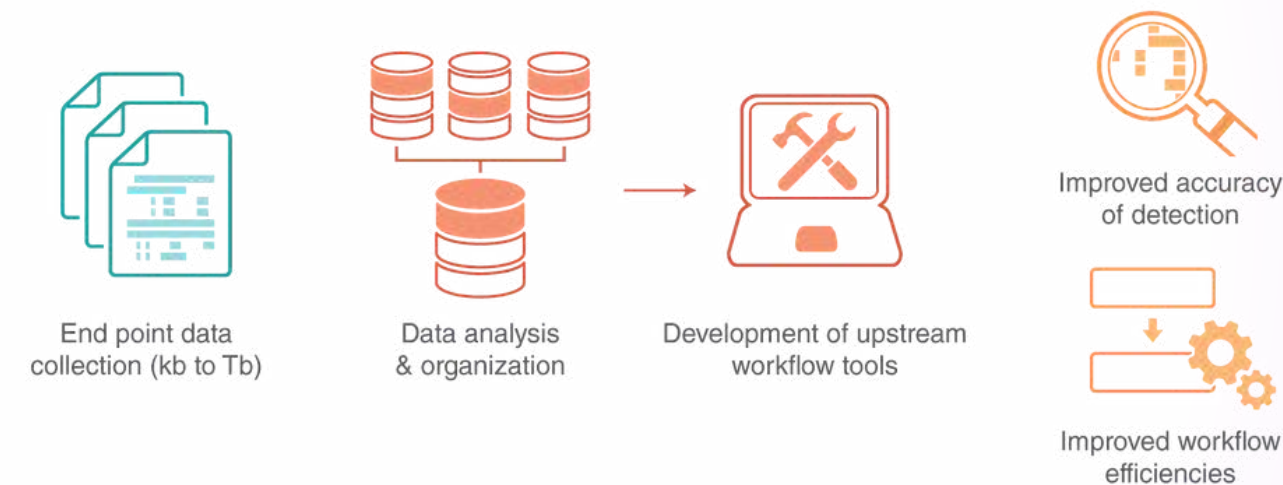
Machine Learning

Some applications of machine learning include medical scans that predict diseases, predicting protein structures, identify cancer related genes, predicting mutation patterns



Computational Biology

Biological data includes DNA sequences, infection rates (as previously mentioned), patient outcomes, gene expression, population health records



Statistics Epidemiology

in

Statistics helps epidemiologists:

- Measure infection trends
- Track mortality rates
- Compare regions
- Predict future cases
- Evaluate interventions

Answers questions like

- Did lockdowns reduce spread?
- Did vaccines reduce hospitalization?
- Which populations are most vulnerable?

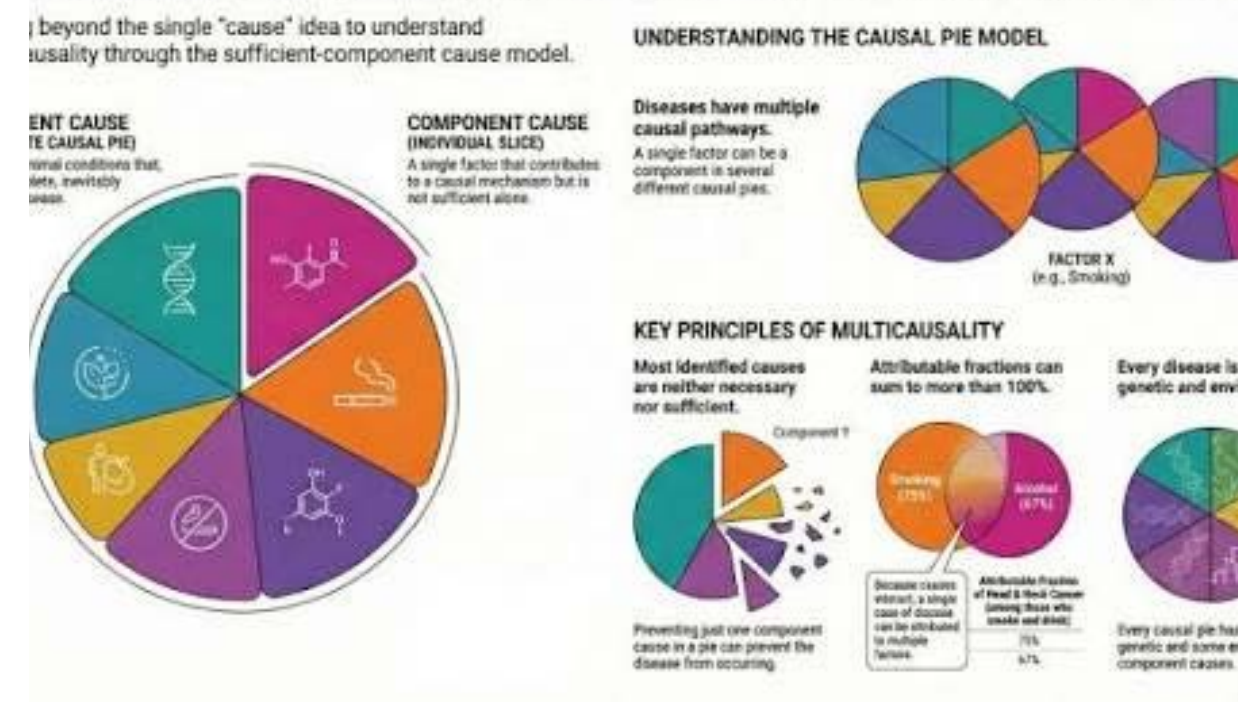
Epidemiology

However because epidemiology is the study of patterns of disease in populations, certain criteria need to be met in order to come to a solid conclusion and prevent confusing correlation with causation.

Bradford Hill criteria - principles used to establish whether an observed statistical association represents a true cause-and-effect relationship rather than just a correlation.



The Causal Pie: A Modern Model for Disease Causation



Rothman Causal Pie - framework visualizing that disease causes are multifactorial, not caused by single factors

Experimental Study

Where participants are exposed to a particular intervention or treatment and are compared to a sample of people that are not exposed with the purpose of understanding more about the intervention itself

Observational Study

An already exposed audience is observed and compared with non-exposed group of people. Used for cases such as viruses in order to not unnecessarily expose people to potential life-threatening scenarios.

Computational Epidemiology

Modeling the future to stop the next pandemic

Definition: using code, math, statistics, and real data to model disease spread.

What data? case counts, contacts, mobility, genomes, vaccines, demographics.

Output: forecasts, intervention tests, risk maps, and public-health decisions.

Recent problems: COVID/Long COVID, avian flu spillover, measles outbreaks, antimicrobial resistance, climate-driven vector disease.

Core idea: build a “digital outbreak” before it happens in real life.

Top 3 formulas / technical core

1) $R_0 = \beta / \gamma$

Average new infections from one infected person in a fully susceptible population.

2) $dS/dt = -\beta SI/N$ | $dI/dt = \beta SI/N - \gamma I$ | $dR/dt = \gamma I$

SIR model: susceptible $\rightarrow$ infected $\rightarrow$ recovered.

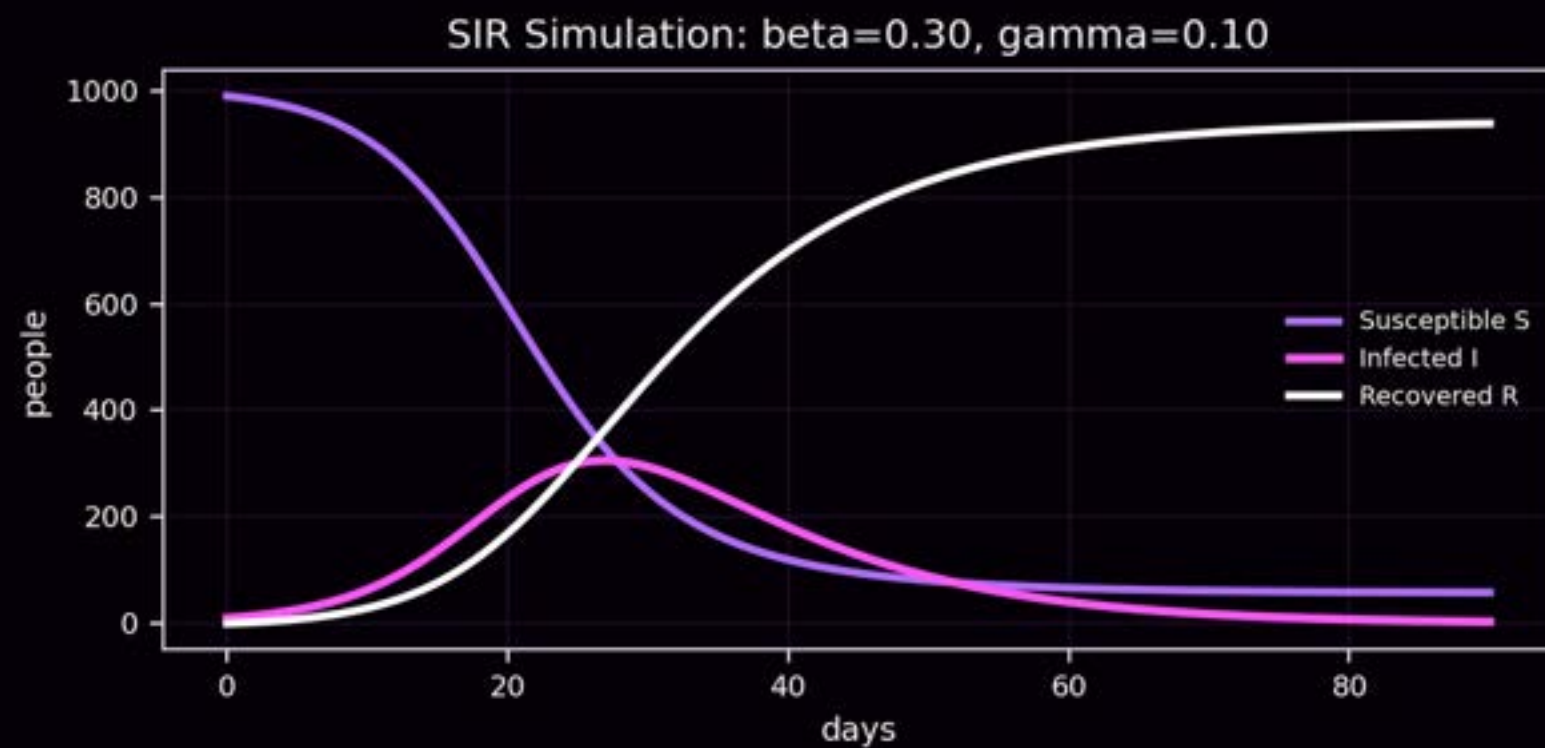
3) $\text{logit}(p) = \ln[p/(1-p)] = \beta_0 + \beta_1 x_1 + \dots$

Regression: estimates risk/probability from variables like age, exposure, vaccine status.

Audience Question: If $R_0 = 3$, what does that mean for one infected person?

Simulation & Code

What people in computational epidemiology actually do



Simulation shown: SIR epidemic curves.
 β = transmission rate; γ = recovery rate.
Try: increase β or lower $\gamma \rightarrow$ bigger, faster outbreak.

Applications / tools used on screen:

Python: NumPy, SciPy, pandas, matplotlib, scikit-learn, networkx, Mesa, Covasim.

R: EpiModel, deSolve, EpiEstim, tidyverse, ggplot2.

Large-scale models: GLEAM/GLEAMviz, agent-based and network models.

```
# tiny SIR update step
N = S + I + R
dS = -beta*S*I/N
dI = beta*S*I/N - gamma*I
dR = gamma*I
S += dS*dt; I += dI*dt; R += dR*dt
```

Mini Problem: If $\beta = 0.42/\text{day}$ and $\gamma = 0.14/\text{day}$, find R_0 . Will the outbreak grow?

Thank You

For Your

Attention

